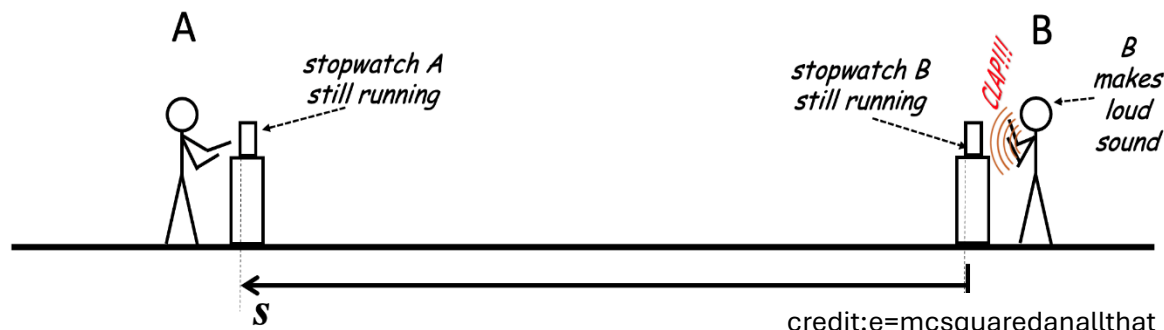




PANAMA



Manual to Lab 5: PHY2048C.

Florida State University

PhyPhox Speed of Sound Experiment

About labs in this class

The labs in this class will have general instructions, and many things need to be figured out by the students. I will be answering any specific questions the students may have without completely giving away the key to the puzzle.

Answer the questions and record your measurements in your lab notebook and then submit the notebook at the end of the activity.

About this lab

In this lab, you will determine if the speed of sound is different inside our classroom vs. outside, and you will attempt to explain this difference, if any.

This lab is different from others because:

- 1) You will be working in pairs.
- 2) You will be using your smartphones:

1st, download the *phyphox* app <https://phyphox.org>.

Second, we will watch this video in class (if you watched it before class, you have had more time to process it):

<https://www.youtube.com/watch?v=uoUm34CnHdE>

Activity 1: Perform the experiment shown in the video in the classroom. You will need to spread out from other groups and take turns so that their claps don't set off your stopwatch. You can go to the storage room and make the experiment there as well. At least three experiments should be able to run at the same time. Make at least 4 trials and take an average. Measure the temperature of the room you are in and record it.

Question 1: Compare your value with the theoretical value given by the Eqn. 39 in the Navier-Stokes handout found in astrosega or Eqn 17.6 in OpenStax. Identify sources of systematic error in this experiment and describe how you control for them (fix them; for instance, make sure a reflected sound wave is not messing up your experiment). The “threshold” setting in Phyphox is very useful to make sure you are recording the main wave pulse of the clap and not something else. *Get your average value within 15% of the theoretical value (if you don't, points will be deducted).* Include Phyphox screenshots of the 4 trials.

Activity 2: Now go outside and perform the experiment there. Compare it with the theoretical prediction, have at most a 15% discrepancy between theory and prediction (fix for any potential source of systematic error). Is there a difference in the value compared to inside? Include Phyphox screenshots of the 4 trials used to calculate the average.

Question 3. Is this the difference you will expect given the equations for the speed of sound? What is the discrepancy of the theoretical vs predicted *difference* in sound speed?

Activity 4: Measure the slope of the plot c_s^2 vs. T using your two values of temperature and your two values of c_s^2 to get the slope. Exchange this slope value with the rest of the class and make a histogram. What is the theoretical value for this slope? Hint: what is c_s^2 equal to given the theoretical formula for the speed of sound?

Question 4: Compare theoretical vs. predicted value for this slope, for the whole class. What is the standard deviation of the whole class for the value of the slope?